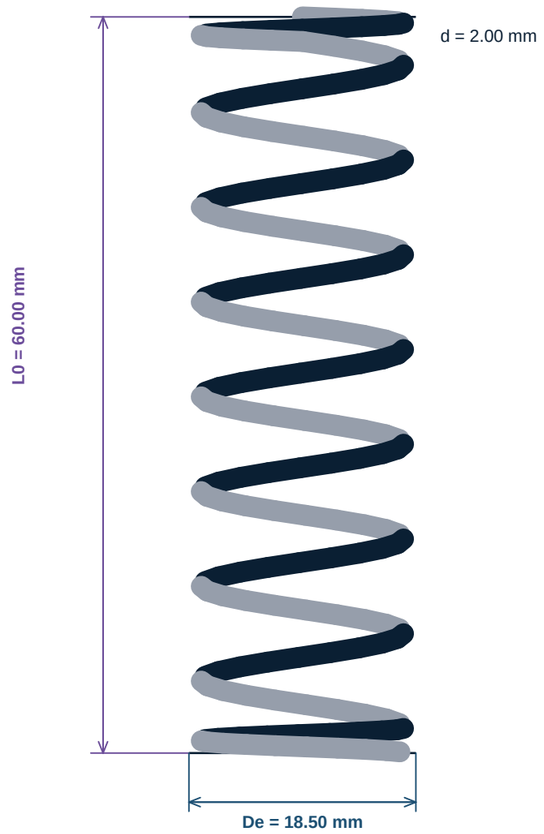
Side view at free length



RH | Closed and ground

1 DIMENSIONS IN MILLIMETRES, INTERPRET PER ASME Y14.5 2 Tolerances on De, Di and D are the same AD from BS EN 15800 Table 3; control one coil diameter. 3 Coil hand RH, default right-hand when not stated.

TABLE 1 - SPRING PARTICULARS — MANUFACTURER

Particular	Sym.	Value
Wire diameter	d	2.000 mm (tolerance per the material standard, EN 10270)
Outside diameter	De	18.500 mm ± 0.45 mm
Inside diameter	Di	14.500 mm ± 0.45 mm
Mean coil diameter	D	16.500 mm ± 0.45 mm (same AD)
Total coils	nt	9.257
Active coils	n	7.257
Free length	L0	60.000 mm - REFERENCE ONLY (EN 15800 6.7)
Load at L1	F1	50.000 N at 50.000 mm ± 7.26 N
Load at L2	F2	125.000 N at 35.000 mm ± 8.38 N
Load at Ln	Fn	176.314 N at 24.737 mm ± 9.15 N - REF, not a specified duty
Coil hand	-	Right-hand (RH)
Ends	-	Closed and ground
Material	-	EN 10270-1 DH
Tensile strength	Rm	1800 N/mm²
Quality grade	-	BS EN 15800 grade 2 (Q = 1.00)
Coil diameter tolerance	AD	± 0.45 mm on De, Di and D
Squareness	e1	<= 3.000 mm (0.050 x L0)
Parallelism	e2	<= 0.555 mm (0.030 x De)
Scope	-	Within the BS EN 15800 normative scope.

TABLE 2 - DESIGN REFERENCE DATA — REFERENCE ONLY

Particular	Sym.	Value
Free length	L0	60.000 mm
Solid length	Lc	18.514 mm
Length at F1	L1	50.000 mm
Length at F2	L2	35.000 mm
Recommended shortest length	Ln	24.737 mm (85 % of available travel)
Deflection at F1	s1	10.000 mm (24.1 % of available travel)
Deflection at F2	s2	25.000 mm (60.3 % of available travel)
Deflection to solid	sc	41.486 mm (100.0 % of available travel)
Deflection at Ln	sn	35.263 mm (85.0 % of available travel)
Corrected stress at L0	tk0	0.0 N/mm²
Corrected stress at L1	tk1	306.4 N/mm²
Corrected stress at L2	tk2	765.9 N/mm²
Corrected stress at Ln	tkn	1080.4 N/mm²
Corrected stress at Lc	tkc	1271.0 N/mm²
Allowable screening stress	tzul	900.0 N/mm² (0.50 x Rm)
Modulus of rigidity	G	81500 N/mm²
Tensile strength	Rm	1800 N/mm²
Spring rate	R	5.0000 N/mm
Mean coil diameter	D	16.500 mm
Spring index	w	8.250
Bergsträsser factor	k	1.1667
Sum of minimum gaps	Sa	6.223 mm (app rule: 0.15 x (L0 - Lc))
Pocket diametral clearance	-	1.500 mm
Approximate wire length	Lw	479.9 mm
Approximate mass	m	11.83 g
Indicative free-length tolerance	AL0	± 1.302 mm indicative (not a limit: two loads given)
Design method	-	BS EN 13906-1

BS EN 15800 grade factor Q, coil factor kf (Annex A.1), force tolerance AF, free-length tolerance AL0, squareness e1, parallelism e2 and the scope limits were checked against published clause values. Coil-diameter tolerances are Table 3 transcribed from a public copy of EN 15800:2008 (not re-read from a licensed PDF). The geometry factor aF is Annex A (A.2) from that same copy; as a cubic in (w/10) it lies within a few percent of published DIN 2095 diagram readings, but the coefficient decimals were not confirmed from a clean print. This is a design aid, not a certified copy of the standard.

HELICAL COMPRESSION SPRING

DRAWING NO. **HC-0200-0185**	DATE **05 Nov 2026**
UNITS **MILLIMETRES (mm)**	SCALE **2:1**
PROJECTION **THIRD ANGLE**	QUALITY **BS EN 15800 GR 2 (Q = 1.00)**
MATERIAL **EN 10270-1 DH**	
GENERAL NOTE **DIMENSIONS IN MILLIMETRES, INTERPRET PER ASME Y14.5**	